Multiplying and Dividing in Scientific Notation

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lessons 2–3 — Multiplying and Dividing in Scientific Notation	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.5

2. Standards

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must. Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not. No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the product is regrouped before it is evaluated, so 'multiply the fronts, add the backs' is read off the structure rather than memorised as a rule.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1, 3 and 7 — one move each, written and up on the cue; dividing in scientific notation was the lowest single figure on the last state report.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 2×10^{10} and 20×10^{9}, which are the same number, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

Essential question. Why must the powers of ten match to add, but not to multiply?

Building on. 4.01 — the coefficient rule and the finishing step; MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.03) — the product and quotient of powers laws.

Working toward. 4.04 — real-world problems and significant digits; MA.912.NSO.1 — operations with rational exponents in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why do the powers of ten have to match before you add, but not before you multiply?	DOK 3
Marcus's power of ten is right and Sara's coefficient is right. Which of the two is closer to the correct answer, and does 'closer' mean anything useful here?	DOK 3
A quotient of two positive numbers comes out as 5.3×10^{-3}. What does the negative exponent tell you about how the two numbers compared?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$(3\times10^5)(3\times10^6)$	***9 × 10¹¹***	9 × 10³⁰ — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; 6 × 10¹¹ — the coefficients were added [8.NSO.1.5: the operation is multiplication]
2	A printer makes 4 × 10³ copies per hour. It runs for 2.5 × 10² hours. Paper costs 3 cents a sheet. How many copies does it make?	***1 × 10⁶***	anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; 10 × 10⁵ — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]
3	$\frac{9.6\times10^8}{1.2\times10^3}$	***8 × 10⁵***	8 × 10¹¹ — the exponents were added [8.NSO.1.5: division subtracts the exponents]; 8.4 × 10⁵ — the coefficients were subtracted instead of divided [8.NSO.1.5]
4	$\frac{3.28\times10^7}{4\times10^{10}}$	***8.2 × 10⁻⁴***	0.82 × 10⁻³ — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies 1 ≤ a < 10]; 8.2 × 10⁻³ — the point moved but the exponent did not
5	$(4 \times 10^7)(2 \times 10^9)$. Which is the product?	***C — 8 × 10¹⁶***	(A) the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication]
6	$(5 \times 10^6)(4 \times 10^3)$. Which is the product, written in scientific notation?	***D — 2 × 10¹⁰***	(A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12 × 10⁹ instead of 1.2 × 10¹⁰]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5]
7	$\frac{7\times10^9}{7\times10^{-6}}$	***1 × 10¹⁵***	1 × 10³ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; 1 × 10⁻¹⁵ — bottom minus top [8.NSO.1.5: top minus bottom]
8	$(8.4 \times 10^7) \div (n \times 10^2) = 4.2 \times 10^5$. What is the value of n?	***n = 2***	4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]; 0.5 — 4.2 ÷ 8.4 instead of 8.4 ÷ 4.2
9	For $(7.2 \times 10^{11}) \div (2.4 \times 10^3)$, Marcus writes 4.8 × 10⁸ and Sara writes 3 × 10¹⁴. Name what each one did wrong, and give the correct quotient.	***3 × 10⁸***	'they both got it wrong' — neither error is named [the item asks what each one did]; 4.8 × 10⁸ — Marcus's error repeated

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give a two-column organiser — COEFFICIENTS on the left, POWERS OF TEN on the right — and have the student split every expression into the two columns before computing anything. The regrouping step in Notes I becomes a physical act, and the two laws are then applied one column at a time.

ELL. Product and quotient are the two nouns, multiply and divide the two verbs. Write them in pairs on the board — PRODUCT goes with MULTIPLY and ADD the exponents, QUOTIENT goes with DIVIDE and SUBTRACT the exponents — and read each board's operation aloud before the boards go up.

Enrichment. Ask for $(4 \times 10^{5})^{2}$ without a calculator (1.6×10^{11}), then for the two numbers in scientific notation whose product is 1.2×10^{7} and whose quotient is 1.2×10^{1} (1.2×10^{4} and 1×10^{3}).

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name Marcus's error and Sara's error separately and give 3×10^{8}. Read the boards, not the papers.

Vocabulary. **product of powers law**; **quotient of powers law**; **regrouping a product**

Materials. Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether 8×10^{16} and 8×10^{63} could possibly both be the answer to whiteboard 5.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 4 test: Multiply numbers written in scientific notation — YZU; Divide numbers written in scientific notation — SGT.